Ailinna Zhang

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EDUCATION

University of Sydney **Expected June 2027**

Bachelor of Advanced Computing (Honours) and Commerce, Data Science, Bioinformatics and Business Analysis

- Honors: Vice Chancellor's Global Mobility Scholarship

University College London **January 2025 – June 2025**

Computer Science, Exchange program

Coursework: Bioinformatics, Computational Modelling for Biomedical Imaging, AI

University of California, San Diego **January 2024 – December 2024**

Computer Science, Exchange program, GPA: 3.763 / 4.0

Coursework: ML, NLP, Algorithms analysis and design, Data Structure and management

Shanghai Jiao Tong University **December 2023– January 2024**

Business Analysis, Summer Exchange program

WORK EXPERIENCES

L.E.K consulting **Medical Consulting Project Intern** **December 2025 – January 2026**

- **Conducted end-to-end landscape research** on a specific oncogenic mutation, synthesizing evidence on current SoC, emerging therapies, and clinical differentiation
- Mapped **patient journey and testing workflow**, identifying barriers affecting diagnosis, referral, and treatment initiation through HCP interviews and desk research
- Assessed how HCPs compare therapies and make treatment choices based on **biomarker eligibility, MOA fit, and perceived clinical value**, surfacing key adoption triggers and objections
- Consolidated insights into a structured storyline and slide deck to **support strategic recommendations on GTM strategy,** including **commercial launch and market access planning**

OC&C **Data Analysis Consulting Project Intern** **October 2025 – January 2026**

- **Automated multilingual product scraping and matching** pipelines in Python, improving cross-market price-comparison accuracy by 70%
- Benchmarked competitor assortments, pricing architecture, and channel strategies (retail, e-commerce, delivery) to **surface region-specific differentiation**
- **Quantified purchase behavior and promotion** uplift to identify drivers of brand loyalty and market-level growth opportunities
- Built Power BI dashboards and translated findings into client-ready recommendations, informing localized pricing and sales actions across global markets

BenHealth **Medical Consulting Intern** **August 2025 – December 2025**

- Researched drug treatment landscapes to **identify clinical gaps and key value drivers** for new therapeutic candidates, highlighting unmet needs and competitive opportunities
- Conducted KOL interviews to map out treatment decision flows, clinical application timing, and patient eligibility, forming the basis for **market opportunity assessments and access assessments**
- **Developed population sizing models and price sensitivity analyses** to evaluate target patient groups and assess commercial acceptance under varying reimbursement and affordability scenarios
- Supported market strategy reports for biopharma clients, **integrating qualitative insights and quantitative forecasting** to inform launch positioning, pricing strategy, and adoption strategies

IPSOS **Strategy3 Consulting Intern** **August 2025 – October 2025**

- **Analyzed biopharma market potential** through KOL interviews and secondary data to identify diagnostic pathways, unmet needs, and reimbursement effects
- **Collected and integrated product and pricing data** from multiple e-commerce platforms to analyse brand performance and cross-country pricing structures, supporting global retail strategy and localization decisions
- Conducted global expert interviews **to assess international manufacturers' localization strategies** for China's market

Merck **Clinical Data Analysis Intern** **June 2025 – September 2025**

- **Developed analytical models for Raman-based experiments**, enhancing component prediction accuracy to 90% and driving a marked improvement in R&D productivity
- **Identified and implemented formulation improvements for client injectables**, reducing viscosity by 20% through targeted component analysis and consistently surpassing quality benchmarks
- Conducted structured **competitor benchmarking** and **market assessment** in upstream bioprocessing, identifying key technology bottlenecks, and new market opportunities to support innovation and strategy

Oliver Wyman **Data Analysis Consulting Project Intern** **September 2024 – April 2025**

- **Evaluated and optimized** credit risk models (PD, LGD), ensuring compliance with Basel III regulations and improving predictive accuracy by 18%
- **Led data collection and stakeholder engagement**, coordinating with multiple bank branches to gather and validate critical loan data across diverse portfolios
- **Diagnosed model weaknesses** and identified new 9 main drivers of default risk out of 30+ factors, to propose and implement data-driven improvements
- **Developed and automated analysis pipelines** using Python and SQL, accelerating model updates and reporting cycles by 30%

AstraZeneca **Business Operations Intern** **July 2024 – September 2024**

- **Supported the launch and evaluation** of major AstraZeneca–hospital partnership projects, strengthening relationships with leading hospitals and medical companies domestically and internationally
- **Coordinated cross-functional teams** to facilitate the implementation of health initiatives, achieving a 95% satisfaction rate from key hospital partners
- **Streamlined communication and collaboration**, integrating feedback and needs from stakeholders to deliver tailored solutions and drive business expansion

RESEARCH and PROJECTS

Stablecoin-Based Cross-Border E-Commerce Payments (Web3, Personal Research)

- Analyse traditional vs stablecoin-based cross-border payment flows for e-commerce sellers, focusing on settlement costs, FX risk and regulation
- Build scenario-based models to compare fees, capital lock-up and P&L under different market and policy scenarios, and assess impact on a representative cross-border seller's unit economics

League of Legend Analysis (Language: Python) **February 2024 – March 2024**

- Led comprehensive data cleaning, missing value imputation, and hypothesis testing to validate gameplay trends and ensure analytical reliability on large-scale match data
- Developed and iteratively optimized predictive models through advanced feature engineering and hyperparameter tuning, achieving an 88% prediction accuracy, significantly outperforming baseline approaches
- Conducted in-depth fairness analysis to identify and mitigate model biases, ensuring equitable performance across different player segments

Decoding Brain Activity for Semantic Video Content Translation (NLP) **April 2024 – December 2024**

- Developing methods to decode semantic video content from brain activity recorded while subjects watch the videos,mapping brain signals to **large language model representations**

Liver Cancer Immune Microenvironment & Therapeutic Modeling (Bioinformatics) **May 2024 - Nov 2024**

- Analyzed pro-tumorigenic immune cell mechanisms in liver cancer microenvironment
- Developed computational models predicting treatment response targeting stromal cells using Python

Predicting Protein Subcellular Localization (Language: Python) **March 2025– May 2025**

- Developed an ensemble machine learning model to predict subcellular localization of eukaryotic proteins based on sequence-derived features, combining Random Forest, Gradient Boosting, and class-specific models
- Engineered and selected 69 key features from protein sequences using Biopython, improving classification accuracy and model interpretability
- Achieved 75% accuracy in blind challenge for protein localization, with 90% high-confidence predictions

Fat–Water Separation in MRI (Bioimaging) **March 2025– May 2025**

- Addressed the challenge of accurately quantifying fat fraction (PDFF) and R2* in MRI by designing and simulating a series of signal models to resolve fat–water ambiguity and model real tissue signals
- Developed visualization tools and a parallelized multi-start optimization pipeline, enhancing parameter estimation robustness under varied tissue types and noise conditions
- Adapted fitting algorithms to accommodate both Gaussian and Rician noise, reducing estimation bias and improving clinical mapping accuracy
- Validated the complete workflow on simulated, phantom, and in vivo datasets, producing fat fraction and R2* maps in strong agreement with reference standards and supporting robust clinical quantification

SKILLS

Technical: Python, Java, HTML, CSS, JavaScript, R, SQL, C, C++, AWS, MATLAB, Git, PowerBI, JMP, Tableau, Stata, Wind, Capital IQ, Industrial research, Bidding document creation

Software: AI Agents and LLM (GPT, Claude, Gemini, Cursor, GitHub Copilot), Microsoft (PPT, Word, Access, Excel), Adobe, Google Workspace, Apple Movie

Languages:

Native: Japanese, Mandarin, English, Shanghainese

Conversational: Spanish, Korean, Cantonese

Basic: Arabic

Hobbies: Piano, Marathon, Swimming, Golf, Dance, Books and Movies

ACTIVITIES

- 1st place, Poesiae Chinese Poetry Recitation Competition
- Perfect Score and High Distinction in Grok Web Coding competition (Intermediate) 2020
- 1st place Badminton Competition
- Badminton Team Captain, Best All-Round player
- 3rd place, Sydney Eisteddfod Senior Singer
- Perfect Score and High Distinction in NCSS Coding Challenge (Intermediate) 2022, (Beginners) 2020
- School representative, IGSA Cross Country Carnival
- IE dance studio dance group captain